

CSA0714 – COMPUTER NETWORKS

ASSIGNMENT

Title

Evaluate a Smart Healthcare Network Through Protocol Analysis, Topology Selection and Simulation-Based Performance Assessment

TEAM MEMBERS

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1. ABSTRACT

Smart healthcare systems depend on reliable and efficient computer networks for communication between medical devices, healthcare professionals, servers, laboratories, pharmacies and emergency services. A failure or delay in healthcare communication can affect the quality and speed of patient care.

This project focuses on the design, simulation and performance evaluation of a smart healthcare network for a multi-floor hospital. The proposed network consists of multiple interconnected zones such as patient monitoring, medical records, laboratory services, pharmacy services and emergency communication. More than 40 end devices are considered, including computers, servers, monitoring systems and other healthcare-related devices.

Different network topologies including star, mesh, bus and hybrid topologies are designed and compared. Network protocols such as TCP, UDP, HTTP, FTP, DNS and ICMP are analyzed under different traffic conditions. The network is simulated using a suitable network simulation platform such as Cisco Packet Tracer, GNS3 or EVE-NG.

The performance of the network is evaluated using response time, bandwidth utilization, throughput, packet delivery ratio, packet loss, jitter and end-to-end delay. Three major scenarios are considered: normal network operation, peak patient-monitoring traffic and network disruption or increased congestion.

The results are used to identify bottlenecks, evaluate protocol behavior, compare topologies and select the most suitable network architecture for a smart healthcare environment. The project demonstrates how network design and protocol selection can improve reliability, scalability and communication efficiency in healthcare systems.

2. PROBLEM STATEMENT

A modern hospital contains multiple departments and healthcare systems that continuously exchange information. Patient monitoring systems generate real-time data, medical records must be accessed quickly, laboratory systems exchange test results, pharmacy systems manage prescriptions and emergency departments require immediate communication. Therefore, a reliable and scalable computer network is required to connect all these systems. The objective of this project is to develop, simulate, validate and assess a smart healthcare communication network for a multi-floor hospital.

The network must contain at least four interconnected network zones and a minimum of 40 end devices, including computers, monitoring systems, servers and other connected healthcare devices. Different network topologies such as star, mesh, bus and hybrid must be constructed and examined. Suitable switches, routers, transmission media and standardized cabling must be selected. The proposed architectures must be evaluated based on reliability, scalability, resource utilization and communication requirements.

The selected architecture must then be implemented using a network simulation or emulation tool such as Packet Tracer, GNS3, EVE-NG or an equivalent platform. The network must generate or observe TCP, UDP, HTTP, DNS, FTP and ICMP traffic under different operating conditions. The performance must be measured under normal operation, peak patient-monitoring traffic and network disruption or increased congestion. The final architecture must be selected based on measured performance and engineering analysis.

3. PROBLEM FORMULATION

The healthcare network can be formulated as a communication system where multiple departments exchange data through wired and wireless network infrastructure.

The major requirements are:

1. Connect at least four healthcare zones.
2. Support a minimum of 40 end devices.
3. Provide communication between departments.
4. Support real-time patient monitoring.
5. Support medical record access.
6. Support laboratory and pharmacy communication.
7. Provide emergency communication.
8. Support multiple network protocols.
9. Maintain reliable communication during congestion.
10. Provide scalability for future expansion.
11. Reduce packet loss and delay.

12. Provide acceptable throughput and response time.
13. Identify possible network bottlenecks.
14. Compare different network topologies.
15. Select the most suitable architecture using measured results.

4. OBJECTIVES

The major objectives of the project are:

- To understand the networking requirements of a smart healthcare environment.
 - To design different network topologies for a hospital environment.
 - To analyze TCP, UDP, HTTP, FTP, DNS and ICMP protocols.
 - To simulate the proposed healthcare network.
 - To generate different types of network traffic.
 - To evaluate network performance under different scenarios.
 - To measure throughput, delay, packet loss and other performance parameters.
 - To identify network bottlenecks.
 - To compare star, mesh, bus and hybrid architectures.
 - To select the most suitable topology.
 - To justify the engineering decisions using measured results.
 - To understand the importance of reliable communication in healthcare networks.
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5. EXPECTED OUTCOMES

After completing the project, the following outcomes are expected:

1. A complete healthcare network architecture will be developed.
2. At least four healthcare network zones will be interconnected.
3. More than 40 end devices will be represented in the network.
4. Star, mesh, bus and hybrid topologies will be evaluated.
5. Different network protocols will be analyzed.
6. Network traffic will be tested under different scenarios.
7. Performance metrics will be collected systematically.
8. Bottlenecks and congestion points will be identified.

9. The most suitable network architecture will be selected.
10. The final design will be justified using simulation results.

6. NETWORK REQUIREMENTS

The proposed hospital network is divided into different logical zones.

Proposed Network Zones

Zone	Main Function	Example Devices
Zone 1 – Patient Monitoring	Monitor patients and collect real-time data	Patient monitoring devices, PCs, monitoring servers
Zone 2 – Medical Records	Store and access patient records	PCs, database servers, printers
Zone 3 – Laboratory	Exchange laboratory reports and test results	Lab PCs, servers, diagnostic systems
Zone 4 – Pharmacy	Manage prescriptions and medicine information	Pharmacy PCs, servers, printers
Zone 5 – Emergency	Support urgent communication	Emergency PCs, monitoring systems, communication devices

The network is designed to support a minimum of 40 end devices.

Example Device Distribution

Network Zone	Number of End Devices
Patient Monitoring	12
Medical Records	10
Laboratory	8
Pharmacy	5
Emergency	5
Total	40

The exact number of devices can be adjusted according to the final simulation implementation.

7. PROPOSED SMART HEALTHCARE NETWORK

The proposed healthcare network consists of multiple departments connected through switches and routers.

The network follows a hierarchical design where end devices connect to access switches, switches connect to higher-level network devices and routers provide communication between different network segments.

Major Components

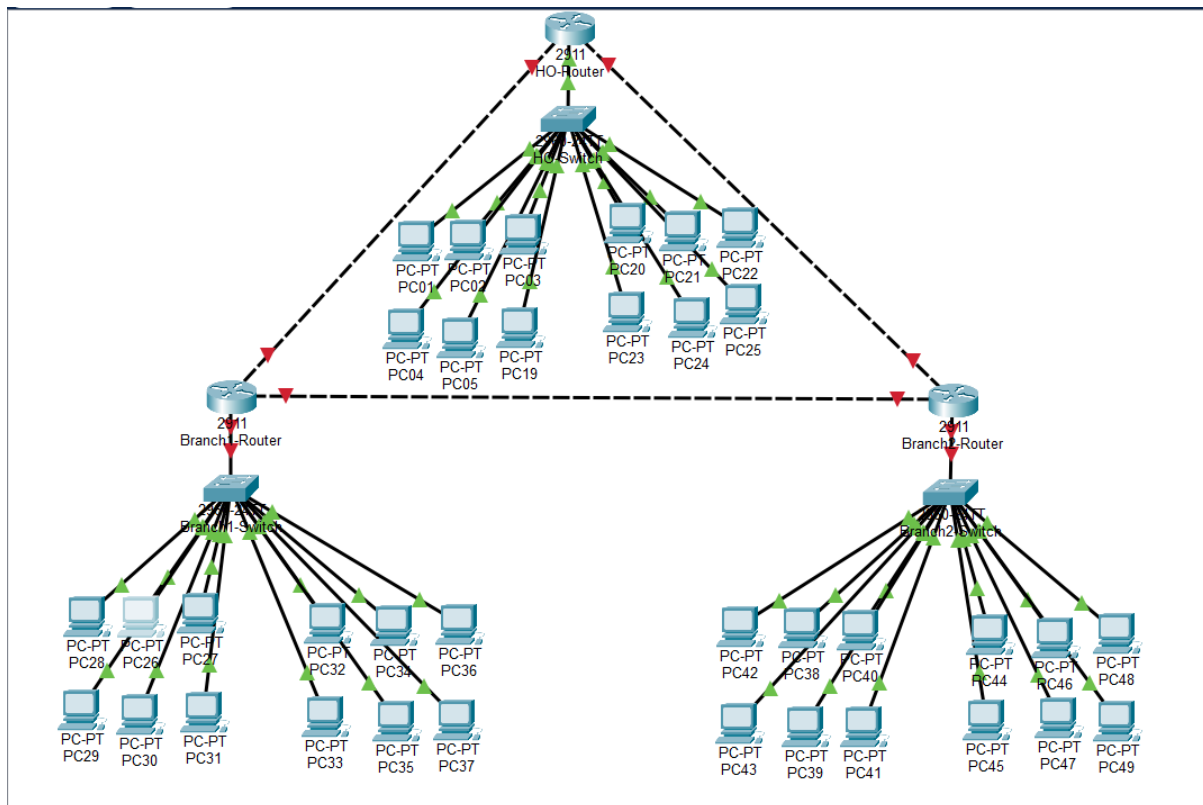
- End-user computers
- Patient monitoring devices
- Servers
- Network switches
- Routers
- Wireless access points where required
- Printers
- Network cables
- Fiber links for backbone connections
- Ethernet connections
- DNS server
- Web server
- File server
- Medical database server

The network is designed to provide reliable communication between departments while allowing future expansion.

8. TOPOLOGY DESIGN

Four major topologies are considered:

1. Star topology
2. Mesh topology
3. Bus topology
4. Hybrid topology
- 5.



8.1 STAR TOPOLOGY

In a star topology, all end devices are connected to a central switch.

Structure

Each computer or healthcare device has a dedicated connection to the central switch.

Advantages

- Easy to install.
- Easy to manage.
- Easy to troubleshoot.
- Failure of one end device does not affect other devices.
- Easy to expand.
- Suitable for departmental networks.

Disadvantages

- Failure of the central switch can affect the entire segment.

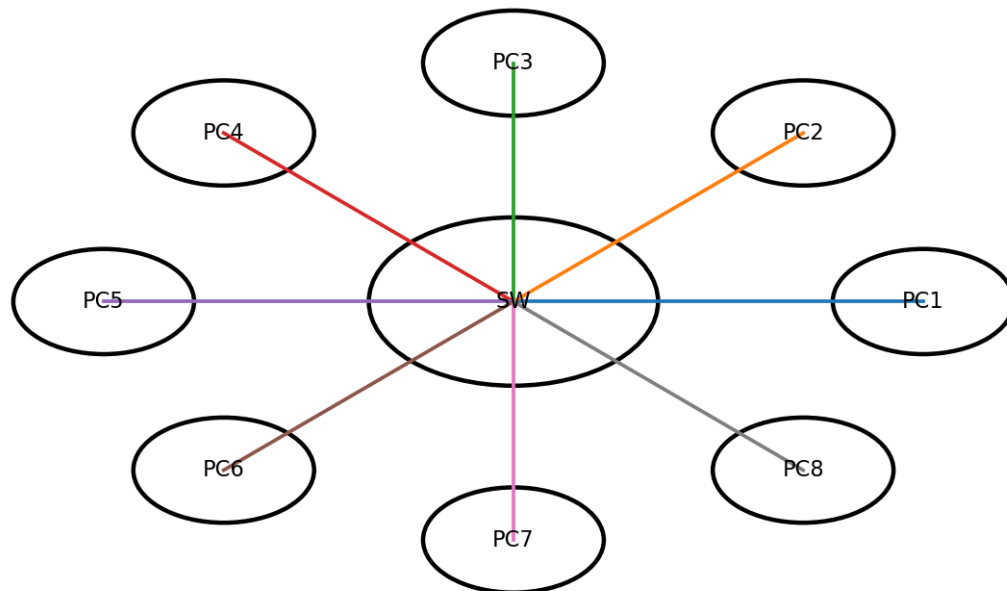
- Requires more cable compared with bus topology.
- Central device becomes an important dependency.

Healthcare Application

Star topology can be used within individual hospital departments such as:

- Medical records
- Pharmacy
- Laboratory
- Administration

Star Topology - Central Switch



8.2 MESH TOPOLOGY

In a mesh topology, devices have multiple connections with other devices.

A full mesh provides multiple paths between network nodes.

Advantages

- Very high reliability.
- Multiple alternative communication paths.
- Better fault tolerance.
- Suitable for critical communication.

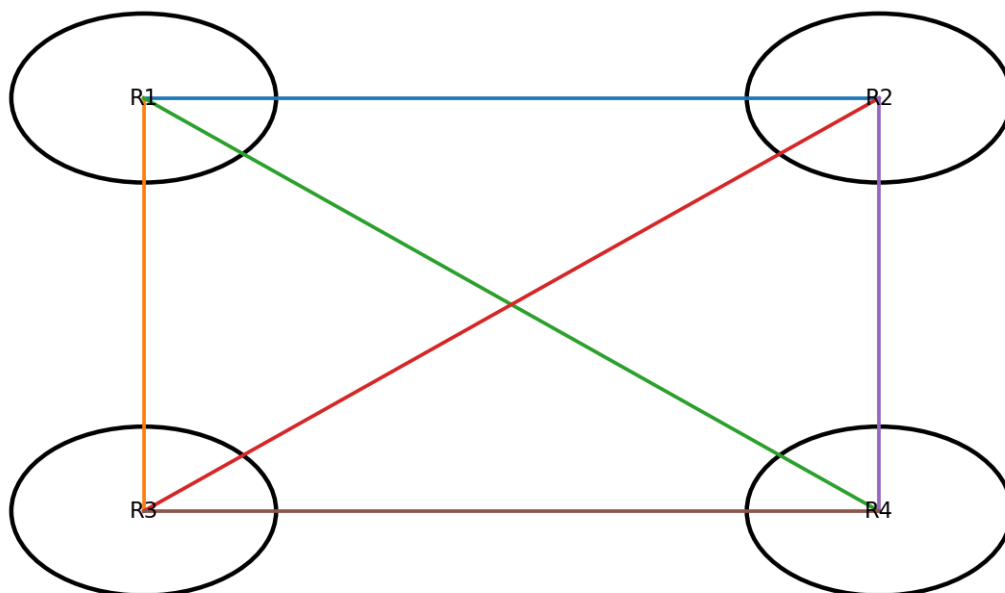
Disadvantages

- High cabling requirement.
- Expensive.
- Difficult to configure and maintain.
- Not practical for connecting every individual end device in a large hospital.

Healthcare Application

Mesh connections can be used for critical backbone or core network links where high availability is required.

Mesh Topology - Multiple Redundant Paths



8.3 BUS TOPOLOGY

In a bus topology, multiple devices share a common communication backbone.

Advantages

- Simple structure.
- Requires less cable.
- Low initial cost.
- Easy to understand.

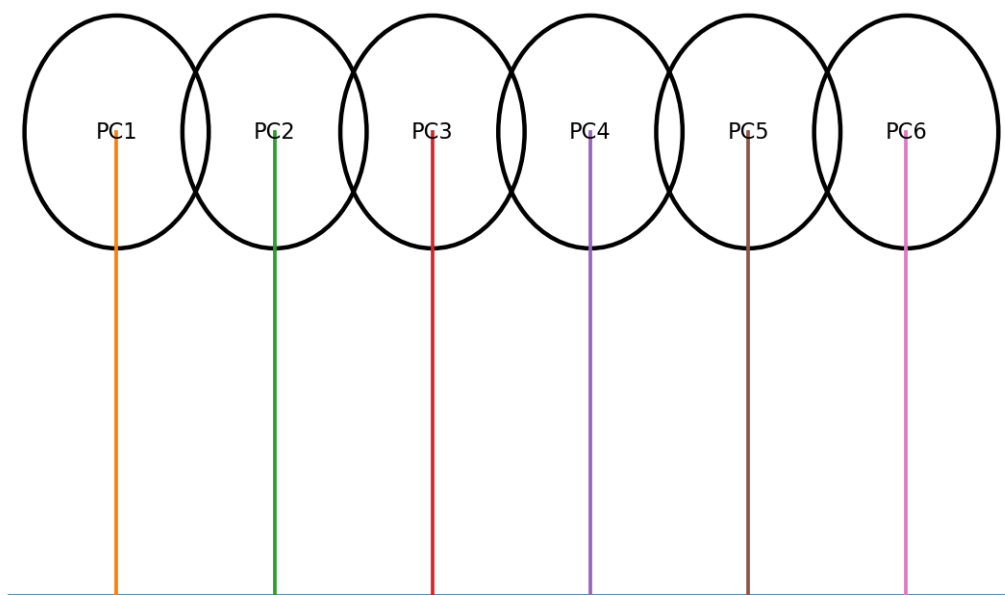
Disadvantages

- Backbone failure can affect many devices.
- Performance decreases when traffic increases.
- Difficult to troubleshoot in large networks.
- Limited scalability.
- Not suitable for modern large healthcare networks.

Healthcare Application

Bus topology can mainly be used as a comparison architecture in the simulation rather than as the final hospital architecture.

Bus Topology - Shared Backbone



8.4 HYBRID TOPOLOGY

A hybrid topology combines two or more different topology types.

For the proposed healthcare network, a combination of star and mesh structures can be used.

Each department can use a star topology, while the major switches or routers can be interconnected using redundant links.

Advantages

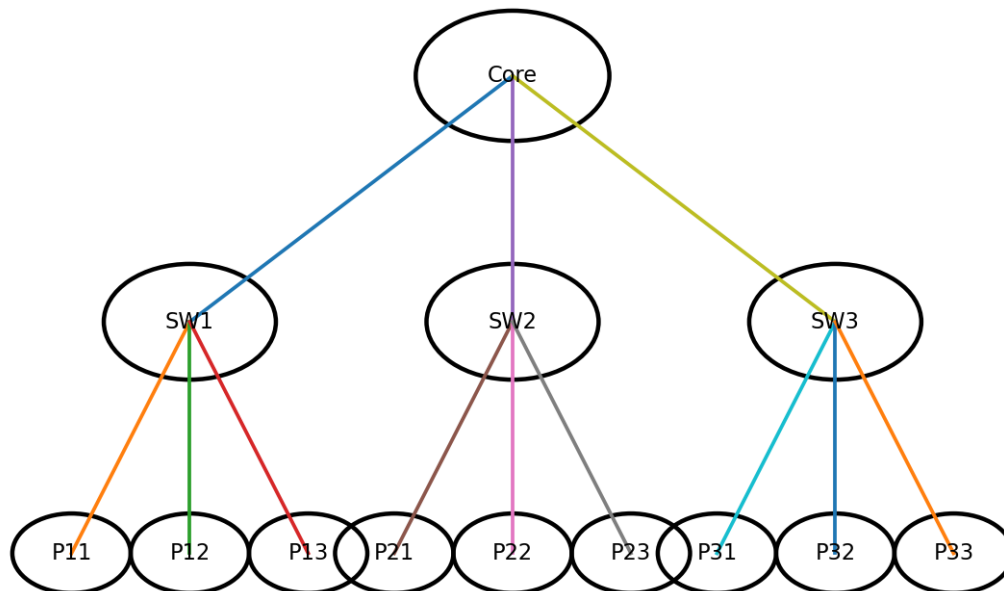
- Highly scalable.
- Flexible.
- Better reliability.
- Supports departmental segmentation.
- Can provide redundancy.
- Suitable for large organizations.
- Easier to expand than a pure mesh network.

Disadvantages

- More expensive than a simple star topology.
- Requires careful planning.
- Configuration is more complex.
- Requires skilled network administration.

Healthcare Application

Hybrid topology is the most suitable architecture for the proposed hospital because different departments have different communication requirements.



9. TOPOLOGY COMPARISON

Parameter	Star	Mesh	Bus	Hybrid
Reliability	High	Very High	Low	Very High
Scalability	High	Low	Low	Very High
Cost	Medium	Very High	Low	Medium/High
Fault Tolerance	Medium	Very High	Low	High
Maintenance	Easy	Difficult	Moderate	Moderate
Traffic Handling	Good	Excellent	Poor under congestion	Excellent
Expansion	Easy	Difficult	Difficult	Easy

Healthcare Suitability High High for backbone Low Very High

Based on the comparison, the hybrid topology provides the best balance between reliability, scalability, performance and cost.

10. NETWORK DEVICES

The following networking devices can be used in the proposed healthcare network.

Device	Purpose
Router	Connects different network segments and provides inter-network communication
Switch	Connects devices within a local network
Server	Provides services such as DNS, HTTP, FTP and database access
Wireless Access Point	Provides wireless connectivity for supported healthcare devices
End Devices	Computers, monitoring systems, printers and other connected devices
Firewall	Provides security and traffic filtering where supported
Network Interface Card	Provides network connectivity to end devices

11. TRANSMISSION MEDIA AND CABLING

Different transmission media are selected according to the network requirements.

Media	Application	Main Advantage
UTPEthernet Cable	End-device connections	Low cost and easy installation
Cat6 Cable	High-speed LAN connections	Supports high-speed communication
Fiber Optic Cable	Backbone connections	High bandwidth and long-distance communication
Wireless	Mobile healthcare devices	Mobility and flexibility

Cabling Standards

Ethernet connections can follow standardized structured cabling practices.

Straight-through Ethernet cables can be used for connections between different types of devices where required, while appropriate switching interfaces can be used for network infrastructure connections.

Fiber optic links are preferred for high-speed backbone connections.

12. PROTOCOL ANALYSIS

The major protocols considered in this project are TCP, UDP, HTTP, FTP, DNS and ICMP.

12.1 TCP

Transmission Control Protocol is a connection-oriented transport layer protocol.

TCP provides reliable communication using mechanisms such as:

- Connection establishment
- Acknowledgement
- Retransmission
- Sequencing
- Flow control
- Congestion control

Healthcare Applications

TCP is suitable for:

- Medical record transfer
- Database communication
- Web applications
- File transfer
- Reliable administrative communication

Advantage

Reliable delivery of data.

Disadvantage

Additional overhead can increase delay compared with UDP

12.2 UDP

User Datagram Protocol is a connectionless transport layer protocol.

UDP does not provide the same reliability mechanisms as TCP.

Healthcare Applications

UDP can be considered for:

- Real-time monitoring
- Time-sensitive data
- Streaming-type communication
- Applications where low latency is more important than retransmission

Advantage

Low protocol overhead and low latency.

Disadvantage

Packets may be lost, duplicated or received out of order.

12.3 HTTP

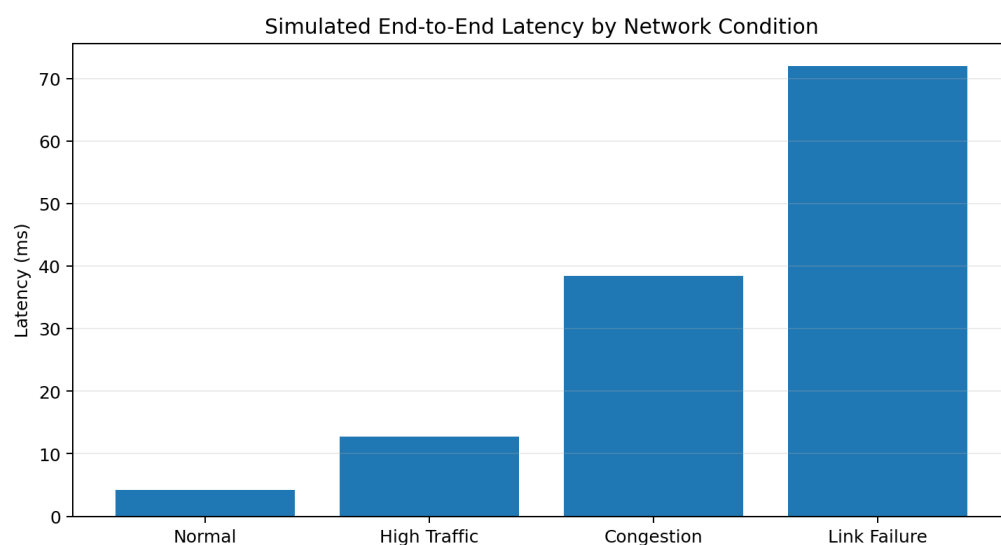
Hypertext Transfer Protocol is an application layer protocol used for web-based communication.

Healthcare Applications

HTTP can be used for:

- Hospital web portals
- Patient record interfaces
- Doctor dashboards
- Laboratory result systems
- Pharmacy applications

HTTP commonly operates over TCP.



12.4 FTP

File Transfer Protocol is used for transferring files between systems.

Healthcare Applications

FTP can represent transfer of:

- Reports
- Documents
- Medical files
- **Administrative data**

Because medical information is sensitive, real-world implementations should use secure alternatives such as encrypted file transfer mechanisms.

12.5 DNS

Domain Name System converts domain names into IP addresses.

For example:

A client may request a healthcare server using a domain name instead of remembering its IP address.

Healthcare Applications

DNS can simplify access to:

- Medical record servers
- Hospital web portals
- Laboratory servers
- Pharmacy systems
- Internal applications

12.6 ICMP

Internet Control Message Protocol is used for network diagnostic and error-reporting purposes.

Healthcare Network Application

ICMP can be used to:

- Test connectivity.
- Identify unreachable devices.
- Measure response time using suitable tools.
- Troubleshoot network problems.

13. PROTOCOL COMPARISON

Protocol	Layer	Connection	Reliability	Typical Use
TCP	Transport	Connection-oriented	High	Medical records, web, file transfer
UDP	Transport	Connectionless	Lower	Real-time/time-sensitive communication
HTTP	Application	Uses TCP	High through TCP	Healthcare web applications
FTP	Application	Uses TCP	High through TCP	File transfer
DNS	Application	Usually UDP for queries; TCP also used when required	Application-dependent	Name resolution
ICMP	Network/Internet	Not a transport connection	Diagnostic protocol	Connectivity and troubleshooting

14. EXPERIMENTAL SCENARIOS

The network is evaluated under at least three scenarios.

SCENARIO 1 – NORMAL OPERATION

In the normal operation scenario, the hospital network operates under normal traffic conditions.

Activities

- Medical staff access patient records.
- Laboratory systems exchange reports.
- Pharmacy systems process prescriptions.
- Patient monitoring devices transmit normal data.
- DNS requests are generated.
- HTTP and FTP communication takes place.

- ICMP packets are used for connectivity testing.

Purpose

To establish the baseline performance of the network.

SCENARIO 2 – PEAK PATIENT-MONITORING TRAFFIC

In this scenario, a large number of patient monitoring devices transmit information simultaneously.

Activities

- Multiple monitoring devices send data.
- Doctors access patient information.
- Laboratory systems continue communication.
- Pharmacy services remain active.
- Network traffic increases significantly.

Purpose

To determine how the network behaves under high traffic conditions.

SCENARIO 3 – NETWORK DISRUPTION / INCREASED CONGESTION

In this scenario, one network link or device is assumed to experience failure or increased congestion.

Activities

- Increase traffic load.
- Introduce a link disruption where supported by the simulator.
- Observe packet loss.
- Observe changes in delay and response time.
- Test alternative communication paths in the hybrid/mesh architecture.

Purpose

To evaluate fault tolerance and network recovery capability.

15.2 BANDWIDTH UTILIZATION

- Bandwidth utilization represents the percentage of available network capacity being used.
- High utilization can indicate heavy network traffic and possible congestion.

15.3 THROUGHPUT

- Throughput represents the amount of useful data successfully transmitted over the network per unit of time.
- Higher throughput generally indicates better data transfer performance.

15.4 PACKET DELIVERY RATIO

Packet Delivery Ratio indicates the percentage of transmitted packets successfully received.

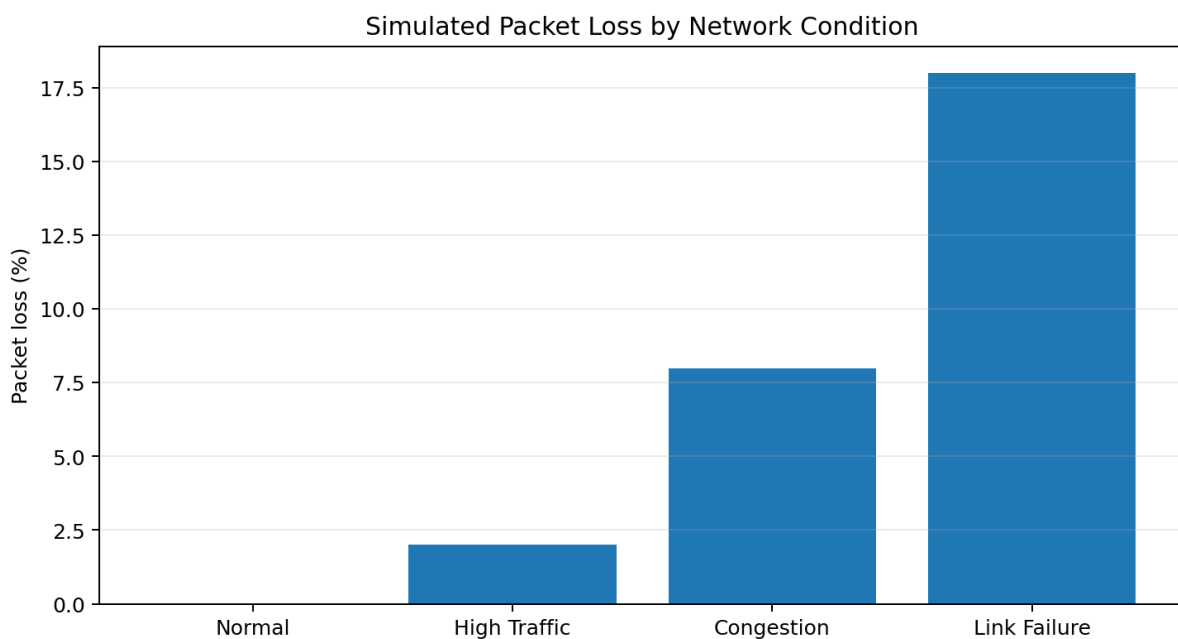
Formula:

$$\text{Packet Delivery Ratio} = (\text{Packets Received} / \text{Packets Sent}) \times 100$$

Higher packet delivery ratio indicates better reliability.

15.5 PACKET LOSS

Packet loss represents the number or percentage of packets that do not successfully reach the destination.



Formula:

$$\text{Packet Loss} = \text{Packets Sent} - \text{Packets Received}$$

High packet loss can negatively affect healthcare communication.

15.6 JITTER

Jitter represents variation in packet delay.

Low jitter is important for real-time communication.

15.7 END-TO-END DELAY

End-to-end delay represents the total time required for data to travel from source to destination.

Lower delay is preferred for time-sensitive healthcare applications.

17. SIMULATION IMPLEMENTATION

The proposed network can be implemented using:

- Cisco Packet Tracer
- GNS3
- EVE-NG
- Any equivalent network simulation/emulation tool

For this project, Cisco Packet Tracer can be used to create the network topology and generate different types of network traffic.

Basic Implementation Steps

1. Create the hospital network zones.
2. Place routers and switches.
3. Add at least 40 end devices.
4. Connect devices using suitable cables.
5. Assign IP addresses.
6. Configure routers and switches.
7. Configure required servers.
8. Configure DNS.
9. Configure HTTP/FTP services where required.
10. Verify connectivity using ICMP.
11. Generate TCP and UDP traffic.
12. Run the normal scenario.
13. Run the peak traffic scenario.
14. Introduce congestion or disruption.
15. Record performance values.
16. Compare the results.
17. Identify bottlenecks.

18. Select the best architecture.

18. BASIC PSEUDOCODE

Pseudocode for Network Performance Evaluation

START

Create hospital network

Create four or more healthcare zones

Add required routers and switches

Add minimum 40 end devices

Configure IP addresses

Configure network connections

Configure required servers

Test connectivity using ICMP

FOR each topology

Create topology

Configure network

Generate normal traffic

Measure performance metrics

Generate peak monitoring traffic

Measure performance metrics

Introduce congestion or link disruption

Measure performance metrics

Record results

END FOR

Compare all topology results

FOR each protocol

Generate appropriate traffic

Measure delay

Measure throughput

Measure packet loss

Measure packet delivery ratio

Measure jitter where applicable

END FOR

Identify bottlenecks

Select the best topology

Validate the selected architecture

Document results

END

19. IMPLEMENTATION CONFIGURATION

The network configuration should include:

IP Addressing

Each department can be assigned a separate logical network.

Zone	Example Network
Patient Monitoring	192.168.10.0/24
Medical Records	192.168.20.0/24
Laboratory	192.168.30.0/24
Pharmacy	192.168.40.0/24
Emergency	192.168.50.0/24
Server Network	192.168.60.0/24

20. SIMULATION VALIDATION

Validation is performed to ensure that the implemented network behaves according to the design requirements.

Validation Activities

1. Verify physical connections.
2. Verify IP addressing.
3. Verify routing.
4. Ping between different network zones.
5. Verify DNS resolution.

6. Verify HTTP communication.
7. Verify FTP communication where configured.
8. Generate TCP traffic.
9. Generate UDP traffic.
10. Record performance measurements.
11. Repeat experiments.
12. Compare expected and observed behavior.

Validation Table

Validation Item	Expected Result	Observed Result
Device connectivity	Successful	To be recorded
Inter-zone communication	Successful	To be recorded
DNS resolution	Successful	To be recorded
HTTP access	Successful	To be recorded
FTP transfer	Successful where configured	To be recorded
TCP communication	Successful	To be recorded
UDP communication	Successful	To be recorded
ICMP connectivity	Successful	To be recorded

21. PERFORMANCE MEASUREMENT RESULTS

The following table should be filled using the actual values obtained from the simulation.

Scenario-Based Results

Metric	Normal Operation	Peak Traffic	Congestion/Disruption
Response Time	To be measured	To be measured	To be measured
Bandwidth Utilization	To be measured	To be measured	To be measured
Throughput	To be measured	To be measured	To be measured
Packet Delivery Ratio	To be measured	To be measured	To be measured
Packet Loss	To be measured	To be measured	To be measured

Jitter	To be measured	To be measured	To be measured
End-to-End Delay	To be measured	To be measured	To be measured

Important: The final report should contain the actual measured values from the simulation rather than assumed values.

22. TOPOLOGY PERFORMANCE COMPARISON

The following table can be completed after running the four topology experiments.

Metric	Star	Mesh	Bus	Hybrid
Response Time	To be measured	To be measured	To be measured	To be measured
Throughput	To be measured	To be measured	To be measured	To be measured
Packet Loss	To be measured	To be measured	To be measured	To be measured
Packet Delivery Ratio	To be measured	To be measured	To be measured	To be measured
Jitter	To be measured	To be measured	To be measured	To be measured
End-to-End Delay	To be measured	To be measured	To be measured	To be measured
Bandwidth Utilization	To be measured	To be measured	To be measured	To be measured

23. PROTOCOL PERFORMANCE ANALYSIS

The protocol performance should be compared based on the actual traffic generated during the simulation.

Protocol	Traffic Type	Expected Behaviour	Important Metrics
TCP	Reliable data	Reliable but higher overhead	Throughput, delay
UDP	Time-sensitive data	Low overhead and low latency	Jitter, packet loss
HTTP	Web communication	Reliable application communication	Response time, throughput
FTP	File transfer	Reliable file transfer	Throughput, response time
DNS	Name resolution	Quick request-response	Response time
ICMP	Diagnostic traffic	Used for connectivity testing	Delay, packet loss

24. BOTTLENECK ANALYSIS

A network bottleneck occurs when a particular network component limits the overall performance of the system.

Possible bottlenecks in the healthcare network include:

- Overloaded switches.
- Limited bandwidth links.
- Excessive UDP traffic.
- High TCP retransmissions.
- Congested backbone links.
- Single points of failure.
- Insufficient server capacity.
- Excessive broadcast traffic.
- Poorly designed network segmentation.

During the simulation, the following indicators should be observed:

1. Increasing response time.
2. Increasing packet loss.
3. Reduced throughput.
4. High bandwidth utilization.
5. Increased jitter.
6. Increased end-to-end delay.

25. TRADE-OFF ANALYSIS

Network design involves several trade-offs.

Cost vs Reliability

A highly redundant network requires additional devices and cables, increasing cost.

However, redundancy improves availability.

Performance vs Complexity

A more complex architecture can provide better performance but requires more configuration and maintenance.

Scalability vs Resource Usage

A scalable architecture requires additional network resources but allows future expansion.

Reliability vs Cost

Mesh connections provide excellent fault tolerance but require significantly more links.

Simplicity vs Fault Tolerance

Star topology is easier to manage but depends heavily on the central switch.

26. ENGINEERING DESIGN JUSTIFICATION

Based on the theoretical comparison, simulation requirements and expected healthcare communication needs, a hybrid architecture is recommended.

The proposed hybrid architecture combines the simplicity of star topology with the reliability of redundant backbone connections.

Each hospital department can use a star arrangement, allowing devices within the department to connect to an access switch. Critical switches or routers can be interconnected using redundant links.

This architecture provides:

- Better scalability.
- Improved reliability.
- Better fault tolerance.
- Easier departmental management.
- Efficient resource utilization.
- Better traffic management.
- Easier future expansion.

For critical patient-monitoring communication, low-delay protocols such as UDP may be considered where occasional packet loss is acceptable and the application itself provides suitable recovery mechanisms.

For medical records, web applications and file transfers where reliable delivery is essential, TCP-based communication is more appropriate.

27. NETWORK SECURITY CONSIDERATIONS

Since healthcare networks handle sensitive information, security must be considered.

Important security measures include:

- User authentication.
- Strong passwords.
- Access control.

- Network segmentation.
- Firewall configuration.
- Secure communication.
- Regular software updates.
- Backup systems.
- Monitoring and logging.
- Restricted administrative access.

Sensitive healthcare information should not be transmitted using insecure protocols in a real-world deployment.

The simulation focuses mainly on network performance, but security is an essential consideration for an actual smart healthcare network.

28. BROADER CONSIDERATIONS AND PROFESSIONAL RESPONSIBILITY

Healthcare networks directly support patient care. Therefore, network engineers must consider reliability, privacy, security and availability.

A network failure may delay access to patient information or monitoring data. Therefore, critical healthcare networks should be designed with redundancy and fault tolerance.

Patient information must also be protected from unauthorized access.

Professional responsibility includes:

- Protecting confidential information.
- Designing reliable infrastructure.
- Testing network failure conditions.
- Documenting configurations.
- Following networking standards.
- Considering scalability.
- Monitoring network performance.
- Avoiding unnecessary risks.

The project demonstrates that technical performance alone is not sufficient. Reliability, security and ethical responsibility are equally important.

29. SDG 3 – GOOD HEALTH AND WELL-BEING

This project is related to Sustainable Development Goal 3 – Good Health and Well-Being.

Smart healthcare networks can support better healthcare services by enabling faster and more reliable communication.

The proposed network can help healthcare organizations by:

- Supporting real-time patient monitoring.
- Improving access to medical records.
- Supporting laboratory communication.
- Improving pharmacy operations.
- Supporting emergency communication.
- Reducing communication delays.
- Improving healthcare system availability.

Therefore, reliable networking infrastructure can contribute to improved healthcare service delivery.

30. COURSE OUTCOME MAPPING

Course Outcome	Project Activity	Achievement
CO3	Analysis of TCP, UDP, HTTP, DNS, FTP and ICMP	Protocol performance is evaluated
CO4	Design of star, mesh, bus and hybrid topologies	Network architectures are created and compared
CO5	Simulation and measurement of network metrics	Performance is measured and analyzed

35. INDIVIDUAL CONTRIBUTION

Jaya Harini B R – 192565031

Contribution:

- Problem analysis.
- Network requirement identification.
- Protocol analysis.
- TCP and UDP comparison.
- HTTP, DNS, FTP and ICMP analysis.
- Experiment and test-plan design.

- Performance metric identification.
- Result interpretation.
- Technical documentation.
- Final report preparation.

Sandiyannittianandam N – 192571029

Contribution:

- Star topology design.
- Mesh topology design.
- Bus topology design.
- Network device selection.
- Transmission media selection.
- Cabling design.
- Network infrastructure planning.
- Packet Tracer topology implementation.
- Connectivity testing.

Yoga Balaji J R – 192572032

Contribution:

- Hybrid topology design.
- Traffic generation.
- Simulation experiment execution.
- Peak traffic testing.
- Congestion/disruption testing.
- Performance result collection.
- Graph preparation.
- Topology comparison.
- Final engineering analysis.

36. MY CONTRIBUTION – JAYA HARINI B R

Working on this project helped me understand how computer networks are applied in a real-world healthcare environment. Before working on this assignment, I mainly understood networking concepts from a theoretical perspective. Through this project, I learned how different network components and protocols work together to support communication between multiple hospital departments.

My major contribution to the project was analyzing the network requirements and studying the behaviour of different communication protocols. I learned the differences between TCP and UDP and understood why protocol selection depends on application requirements. TCP is suitable for applications where reliable delivery is important, while UDP can be useful for time-sensitive communication where lower delay is important. I also studied HTTP, DNS, FTP and ICMP and understood their roles in a healthcare network.

Another important learning experience was understanding performance metrics. I learned how throughput, response time, packet loss, packet delivery ratio, jitter, bandwidth utilization and end-to-end delay can be used to evaluate a network. These metrics help engineers determine whether a network is performing efficiently or whether congestion and bottlenecks are present.

The project also improved my understanding of network topology. Comparing star, mesh, bus and hybrid architectures helped me understand that there is no single topology suitable for every situation. A healthcare environment requires reliability, scalability and fault tolerance, which makes topology selection an important engineering decision.

I also learned the importance of documentation and systematic testing. Instead of simply creating a network, the network must be tested under different conditions and the results must be recorded and analyzed.

Overall, this project improved my knowledge of computer networks and helped me understand how theoretical concepts can be applied to a practical healthcare communication system. It also improved my teamwork, technical documentation and problem-solving skills.

39. LIMITATIONS

The project has the following limitations:

1. Simulation results may not exactly represent a real hospital network.
2. Packet Tracer may not support every advanced networking feature.
3. Real healthcare devices may have different traffic characteristics.
4. Security testing is limited compared with a real healthcare deployment.
5. The performance results depend on the simulation configuration.
6. Wireless performance may vary in a real hospital due to physical interference.
7. Hardware limitations are not fully represented in a simulation environment.

40. FUTURE ENHANCEMENTS

The proposed system can be improved in the future by adding:

- Advanced network security.
- Firewall and intrusion detection systems.
- Wireless healthcare devices.
- IoT-based patient monitoring.
- Cloud-based healthcare services.
- Software-defined networking.
- Network automation.

- Artificial intelligence for traffic prediction.
- Automatic congestion management.
- Redundant servers.
- High-availability routing.
- Real-time network monitoring.
- Quality of Service mechanisms.
- Secure encrypted communication.

AI-based network management can also be introduced to predict congestion and automatically select efficient communication paths.

41. ENGINEERING RECOMMENDATION

Based on the theoretical analysis, a hybrid topology is recommended for the proposed smart healthcare network.

The recommended design consists of:

- Star topology within departments.
- Redundant connections between important network devices.
- Switches for local connectivity.
- Routers for inter-network communication.
- Fiber optic backbone where high bandwidth is required.
- Ethernet connections for end devices.
- Dedicated servers for healthcare applications.
- Suitable network segmentation.
- Monitoring and fault-detection mechanisms.

The final selection must be supported by the actual simulation results.

42. CONCLUSION

This project evaluated the design and performance of a smart healthcare network using protocol analysis, topology comparison and simulation-based performance assessment. The proposed network was designed to support multiple hospital departments including patient monitoring, medical records, laboratory services, pharmacy services and emergency communication.

Star, mesh, bus and hybrid topologies were studied and compared based on reliability, scalability, cost, fault tolerance and performance. Protocols such as TCP, UDP, HTTP, FTP, DNS and ICMP were analyzed according to their communication characteristics and healthcare applications. The network was evaluated under normal operation, peak patient-monitoring

traffic and congestion or network disruption. Performance metrics including response time, bandwidth utilization, throughput, packet delivery ratio, packet loss, jitter and end-to-end delay were selected for evaluation.

The project demonstrates that healthcare networks require more than basic connectivity. They must provide reliable, scalable and efficient communication while considering security and professional responsibility.

A hybrid architecture is recommended because it can combine departmental star networks with reliable backbone connections. However, the final architecture should be selected using the actual measured simulation results. Overall, the project provides practical knowledge of computer network design, protocol analysis, topology selection, simulation and performance evaluation while demonstrating the importance of reliable communication in smart healthcare systems.

43. REFERENCES

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To apply subnetting and VLSM techniques.

To assign suitable default gateways.

To configure routing between different network segments.

To establish communication between the Head Office and Branch Offices.

To design and compare star, mesh, bus, and hybrid topologies.

To select appropriate routers, switches, end devices, and transmission media.

To select suitable cabling standards for the network.

To configure and analyze TCP communication.

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